BIOMEDICAL LIMITED ILORIN	STANDARD TESTING PROCEDURE **TITLE:** STANDARD CHEMICAL TESTING PROCEDURE FOR BIOMEDICAL PARACETAMOL SYRUP (60ml)		
DEPARTMENT:	**QUALITY**	**SECTION:**	**QUALITY CONTROL**
STP NO.:	**BML/QA/STP/FPS/001/F01**	**PAGE NO.:**	**Page 1 of 7**
STP Ref.:	**BML-QDG-QA-105**	**REF. BOOK**	**BP 2024/IN-HOUSE**
EFFECTIVE DATE:	**04/01/2024**	**REVISION NO:**	**01**
REVIEW DATE:	**04/01/2026**	**SUPERSEDE:**	**00**

S/N	TEST	DESCRIPTIONS OF PROCEDURE
1.	**pH**	1. Take sample sufficient to immerse the electrode-head, and dip the electrode in the sample. 2. Wait until the reading stabilises, then take readings.
2.	**Paracetamol** **120mg/5ml**	**METHOD 1:** ***Standard solution***: 10mcg/mL in 0.1M hydrochloric acid. • Weigh accurately and dissolve 0.0249g of standard paracetamol in 0.1M HCl, and make up the volume to 50mL • From the resulting solution, take 1mL to 50mL with the same solvent. ***Sample solution***: • Dissolve an equivalent of 0.5g PCM [20ml Syrup] in 0.1M HCl and make up to 100ml std. flask. • Dilute 1ml of resulting solution with 0.1M HCl to 50ml std. Flask. • Take 5ml of final solution and dilute to 50 ml STD flask. • Read absorbance at 244nm and compare results with standard **% Assay of Paracetamol** = $\frac{absorbance of sample}{absorbance of standard}$ x 100

	DESIGNATION	SIGNATURE	DATE
PREPARED BY:	**QUALITY CONTROL SENIOR ANALYST**		
REVIEWED BY:	**QUALITY CONTROL HEAD**		
APPROVED BY:	**QUALITY ASSURANCE MANAGER**		

REFERENCE: Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 2

METHOD 2:

Standard Solution:

1mg/100mL

- Weigh accurately and dissolve 120mg of standard paracetamol in 0.1M HCl, and make up the volume to 100mL
- From the resulting solution, take 1mL to 100mL with the same solvent.

Sample Solution:

- Dissolve an equivalent of 120mg of the syrup (5mL) of the syrup make up the volume to 100mL with 0.1M HCl.
- From the resulting solution, take 1mL to 100mL with the same solvent
- Read absorbance at 244nm and compare results with standard

$$\textbf{\% Assay of Paracetamol} = \frac{absorbance of sample}{absorbance of standard} \times 100$$

NOTE: in acidic solution, paracetamol has lambda max 246nm, whereas in alkaline solution, it has lambda max 257nm. This bathochromic shift 246→257 nm is due to

molecular re-arrangement in the solvent.

REFERENCE:

Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 2

METHOD 3:

Standard Solution:

1mg/100mL

- Weigh accurately and dissolve 100mg of standard paracetamol in 0.1M NaOH, and make up the volume to 100mL
- From the resulting solution, take 1mL to 100mL with the same solvent.

Sample Solution:

- Dissolve an equivalent of 100mg of the syrup (4mL) of the syrup in 0.1mL and make up the volume to 100mL.
- From the resulting solution, take 1mL to 100mL with the same solvent
- Read absorbance at 257nm and compare results with standard

% Assay of Paracetamol = $\frac{absorbance of sample}{absorbance of standard}$ x 100

		NOTE: in acidic solution, paracetamol has lambda max 246nm, whereas in alkaline solution, it has lambda max 257nm. This bathochromic shift 246→257 nm is due to molecular re-arrangement in the solvent. **REFERENCE:** Sethi's Quantitative Analysis of Pharmaceutical Formulations. Vol. 2
3.	**Microbial Enumeration Test**	- Determination of total aerobic microbial count (TAMC) and total combined yeasts and molds count (TYMC) by the pour plate method. - Dispense 1 ml of diluted sample in the center of the Petri plate using a sterile micropipette or calibrated pipette. - Open the lid of the prepared media, flame its mouth. - Pour about 15 mL of sterilized molten media at the appropriate temperature above the sample. - Close the lid of the plate then mix the sample in the media properly by gently swirling the plate. The plate is generally swirled in an "S" or "8" shape. - Allow the media to completely solidify. - Incubate the plate in an inverted position under suitable incubation conditions (Incubate the plates of Soybean Casein Digest Agar at 30°C-35°C for 3-5days for TAMC and the plates of Sabouraud Dextrose Agar at 20°C to 25°C for 5-7 days.

TEST FOR ABSENCE OF PATHOGEN

- Measure 10ml /10g of the samples, dilute 1n 90ml of nutrient broth
- Transfer 10ml into 90ml of Tryptone soya broth
- Incubate at 30-35^{0}c for 18-24hours
- Transfer 1 ml of the dilution into 100ml of MacConkey broth and incubate at 42-44°c in 18-72hours
- Subculture on MacConkey Agar or Eosin Methylene blue Agar and incubate at 30-35°c

- For negative control, proceeding same as above except without sample, streak a portion of incubated sterile MacConkey broth medium on the surface of Eosin Methylene Blue Agar.
- Incubate plates at 30 to 35° for 48 hours.

A. Result Interpretation

- Following the incubation, developed colonies are observed. Each colony will account for one viable microbial cell or one colony-forming unit.
- If all the colonies are of the same type, then we can predict that the sample contained only one type of microbial genera.
- However, if there are different genera or species producing similar types of colonies, it

is required to carry out identification test.

- If the colonies are of different morphology, then we can conclude that the sample contained a mixed population.

Count the colonies and calculate CFU/mL by using the formula:

$$\text{CFU/mL} = \frac{\text{Total number of colonies obtained x dilution factor}}{\text{Volume of sample used (aliquot)}}$$

- This will give the total number of viable microbial cells present in the given sample.
- For optimum count, the number of colonies must be between 20 – 200 CFU/mL. Beyond this limit, the whole procedure must be repeated.
- If the number of the colony is less than 20, it is suggested to use the sample of lower dilution, whereas, if the total number of colonies exceeds 200, it is suggested to use the sample of higher dilution on successive repeats.
- The result should be recorded following the Standard testing procedure SOP **BML-QDG-QA-105**
- If the colonies are fused or the whole plate is covered with a single colony, then report as “too numerous to count” (TNTC) and repeat the process of taking the sample at a higher dilution.

CURRENT REVISION	SUPERSEDE	REASON FOR REVIEW
00	NIL	FIRST ISSUE
01	00	Review of the procedures Change in header and footer format

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PREPARED BY:	QUALITY CONTROL SENIOR ANALYST		
REVIEWED BY:	QUALITY CONTROL HEAD		
APPROVED BY:	QUALITY ASSURANCE MANAGER		